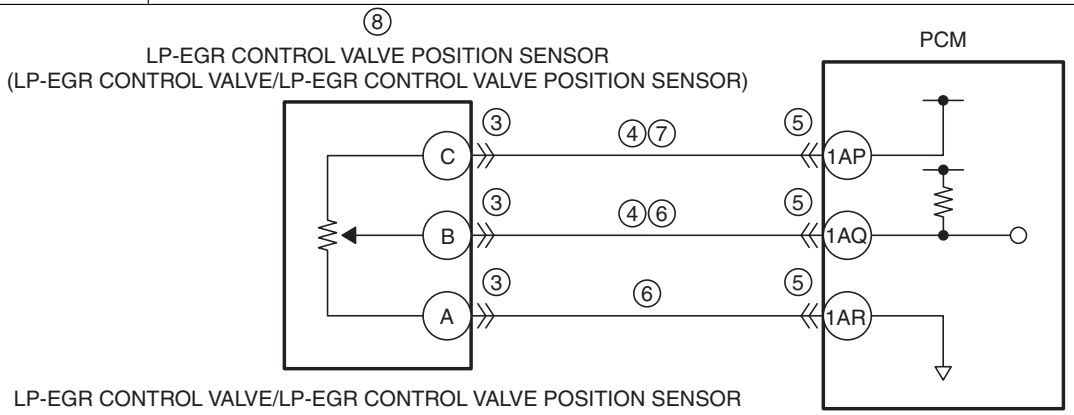
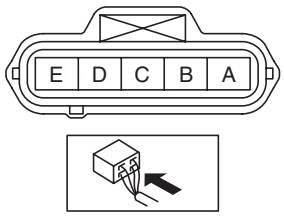
DTC P0407:00 [SKYACTIV-D 1.5]

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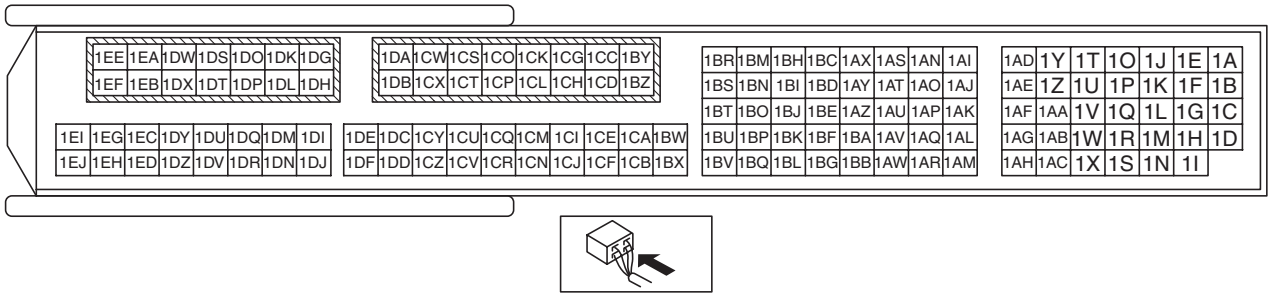
DTC P0407:00	LP-EGR control valve position sensor circuit low input
DETECTION CONDITION	• If the PCM detects the LP-EGR valve position sensor voltage at the PCM terminal 1AQ is **less than 0.24 V** for **2 s** with the following condition met, the PCM determines that the LP-EGR valve position sensor circuit voltage is low. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) **Diagnostic support note** • This is a continuous monitor (EGR system). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque. • Inhibits the EGR control.
POSSIBLE CAUSE	• LP-EGR control valve/LP-EGR control valve position sensor connector or terminals malfunction • Short to ground in wiring harness between the following terminals: — LP-EGR control valve/LP-EGR control valve position sensor terminal C—PCM terminal 1AP — LP-EGR control valve/LP-EGR control valve position sensor terminal B—PCM terminal 1AQ • PCM connector or terminals malfunction • LP-EGR control valve position sensor signal circuit and ground circuit are shorted to each other • Open circuit in wiring harness between LP-EGR control valve/LP-EGR control valve position sensor terminal C and PCM terminal 1AP • LP-EGR control valve position sensor malfunction • PCM malfunction



LP-EGR CONTROL VALVE/LP-EGR CONTROL VALVE POSITION SENSOR WIRING HARNESS-SIDE CONNECTOR



PCM WIRING HARNESS-SIDE CONNECTOR



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS WHEN DTC WAS DETECTED TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture of the PC function. • Record the freeze frame data/snap shot data.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT LP-EGR CONTROL VALVE/LP-EGR CONTROL VALVE POSITION SENSOR CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the LP-EGR control valve/LP-EGR control valve position sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
4	**INSPECT LP-EGR CONTROL VALVE POSITION SENSOR CIRCUIT FOR SHORT TO GROUND** • Verify that the LP-EGR control valve/LP-EGR control valve position sensor connector is disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — LP-EGR control valve/LP-EGR control valve position sensor terminal B — LP-EGR control valve/LP-EGR control valve position sensor terminal C • Is there continuity?	Yes	If the short to ground circuit could be detected in the wiring harness: • Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: — LP-EGR control valve/LP-EGR control valve position sensor terminal C—PCM terminal 1AP — LP-EGR control valve/LP-EGR control valve position sensor terminal B—PCM terminal 1AQ **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. If the short to ground circuit could not be detected in the wiring harness: • Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 9.
		No	Go to the next step.
5	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.

STEP	INSPECTION		ACTION
6	**INSPECT LP-EGR CONTROL VALVE POSITION SENSOR SIGNAL CIRCUIT AND GROUND CIRCUIT FOR SHORT TO EACH OTHER** • Verify that the LP-EGR control valve/LP-EGR control valve position sensor and PCM connectors are disconnected. • Inspect for continuity between LP-EGR control valve/LP-EGR control valve position sensor terminals A and B (wiring harness-side). • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • LP-EGR control valve/LP-EGR control valve position sensor terminal B—PCM terminal 1AQ • LP-EGR control valve/LP-EGR control valve position sensor terminal A—PCM terminal 1AR **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to each other. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to each other. Go to Step 9.
		No	Go to the next step.
7	**INSPECT LP-EGR CONTROL VALVE POSITION SENSOR POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT** • Verify that the LP-EGR control valve/LP-EGR control valve position sensor and PCM connectors are disconnected. • Inspect for continuity between LP-EGR control valve/LP-EGR control valve position sensor terminal C (wiring harness-side) and PCM terminal 1AP (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between LP-EGR control valve/LP-EGR control valve position sensor terminal C and PCM terminal 1AP. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.
8	**INSPECT LP-EGR CONTROL VALVE POSITION SENSOR** • Reconnect all disconnected connectors. • Inspect the LP-EGR control valve position sensor. (See LP-EGR CONTROL VALVE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction**?**	Yes	Replace the LP-EGR control valve, then go to the next step. (See LP-EGR CONTROL VALVE REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
9	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the KOEO or KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
10	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.